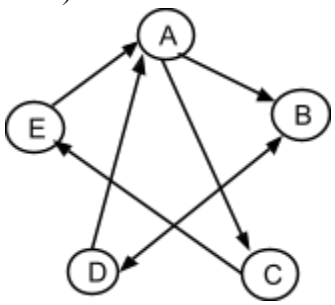
American Computer Science League

2023-2024 ● Contest 4: Short Problems ● Junior Division

1. Graph Theory

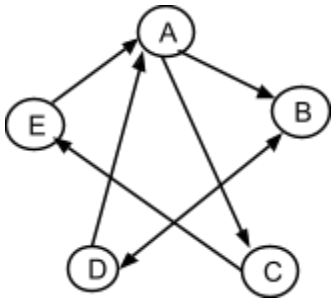
How many ordered pairs of vertices do not have paths of length 2 in the following directed graph? For example, (B,A) means from B to A. (This is the same graph as #2.)



A. 12
B. 14
C. 15
D. 16
E. 20

2. Graph Theory

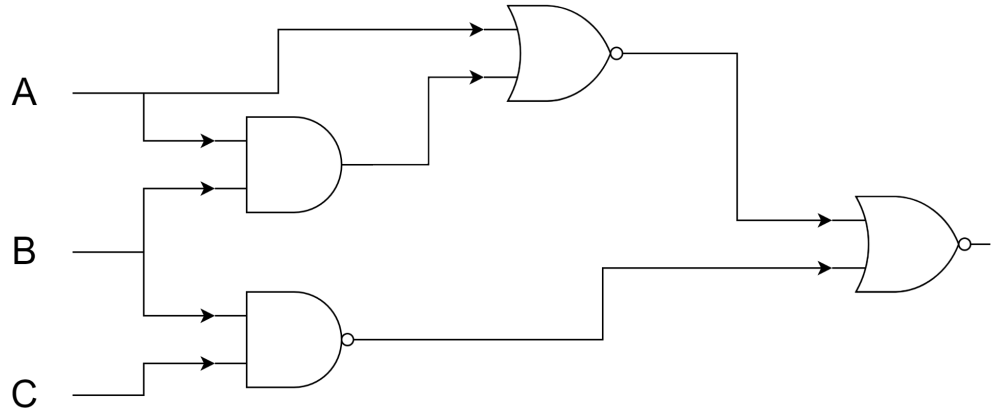
How many cycles are there in the directed graph below? (This is the same graph as #1.)



A. 2
B. 3
C. 4
D. 5
E. 6

3. Digital Electronics

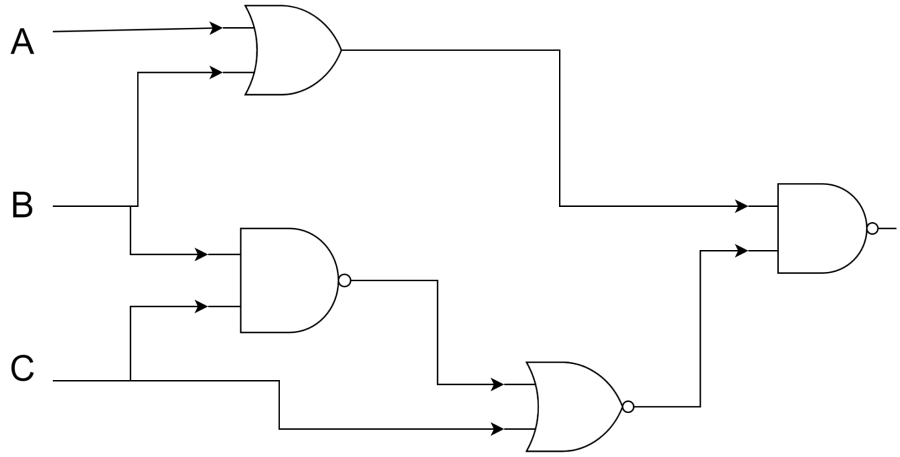
Simplify the Boolean expression represented by the following digital circuit:



A. $A + B + C$
B. $AB + C$
C. $A + BC$
D. ABC
E. BC

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

5. What Does This Program Do? (Strings)

Define the functions val(x) to get the numeric value of a string and str(x) to make a numeric value into a string. For example, val("9") = 9 and str(9) = "9". What will be the output for the following program?

```
s = "639284510327493"
ans = ""
for x = 0 to len(s)-1
     v = val(s[x])
     if v < 5 then
          v = v * 2
     else
          v = int(v / 2)
     end if
     ans = str(v) + ans
next x
output ans
```

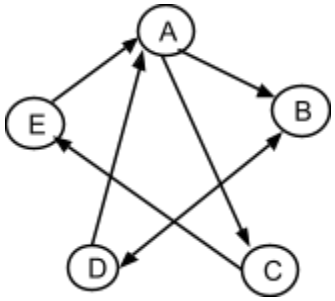
A. 648346022844463
B. 6483460210844463
C. 142311002241413
D. 364448220643846
E. 658446023824563

American Computer Science League

2023-2024 ● Contest 4: Short Problems ● Junior Division

1. 图论

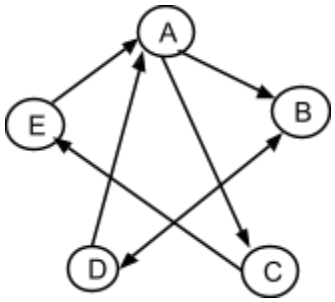
请问在**下述有向图中，有多少个有序顶点对之间没有长度为** 2 **的路径？比如，**(B, A) 表示**从** B 到 A。(下图**与第** 2 **题的图相同。**)



A. 12
B. 14
C. 15
D. 16
E. 20

2. 图论

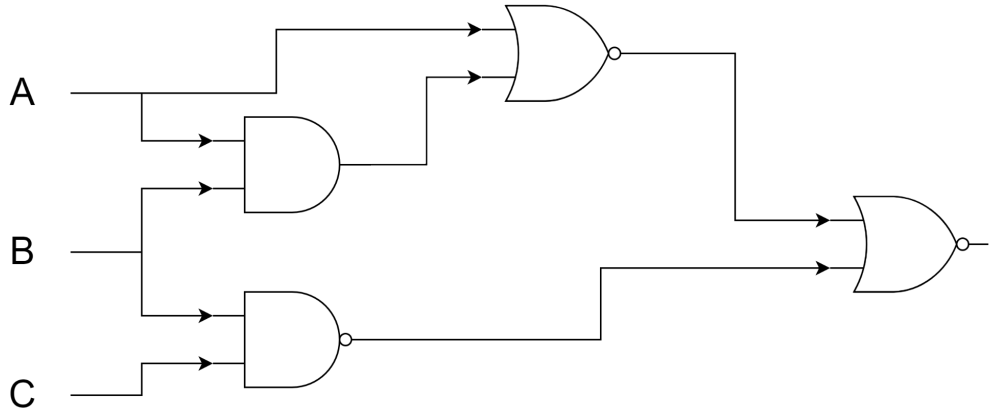
请问**在下述有向图中有多少个环？**(下图**与第** 1 **题中的图相同。**)



A. 2
B. 3
C. 4
D. 5
E. 6

3. 数字电路

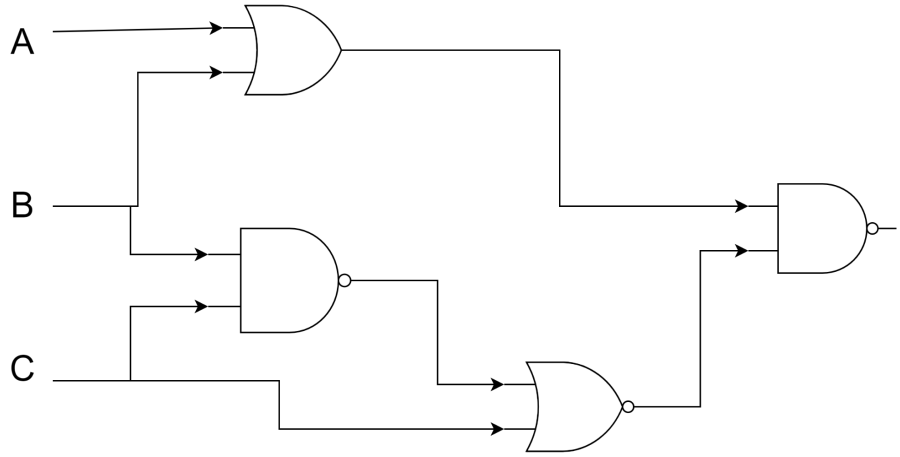
简化**下列用数字电路表示的布尔表达式：**



A. $A + B + C$
B. $AB + C$
C. $A + BC$
D. ABC
E. BC

4. 数字电路

请问**有多少个有序三元组可以使得下列数字电路为真？**



A. 0
B. 2
C. 4
D. 6
E. 8

5. 这个程序是做什么的？(字符串)

定义**函数** val(x) **用以获取字符串的数值；**定义**函数** str(x) **将数值转换为字符串。**例如, val("9") = 9 、 str(9) ="9"。**请问**下列**程序会输出什么**?

```
s = "639284510327493"
ans = ""
for x = 0 to len(s)-1
     v = val(s[x])
     if v < 5 then
          v = v * 2
     else
          v = int(v / 2)
     end if
     ans = str(v) + ans
next x
output ans
```

A. 648346022844463
B. 6483460210844463
C. 142311002241413
D. 364448220643846
E. 658446023824563